

ADR 0115 — A mutation proof cannot see what it does not run

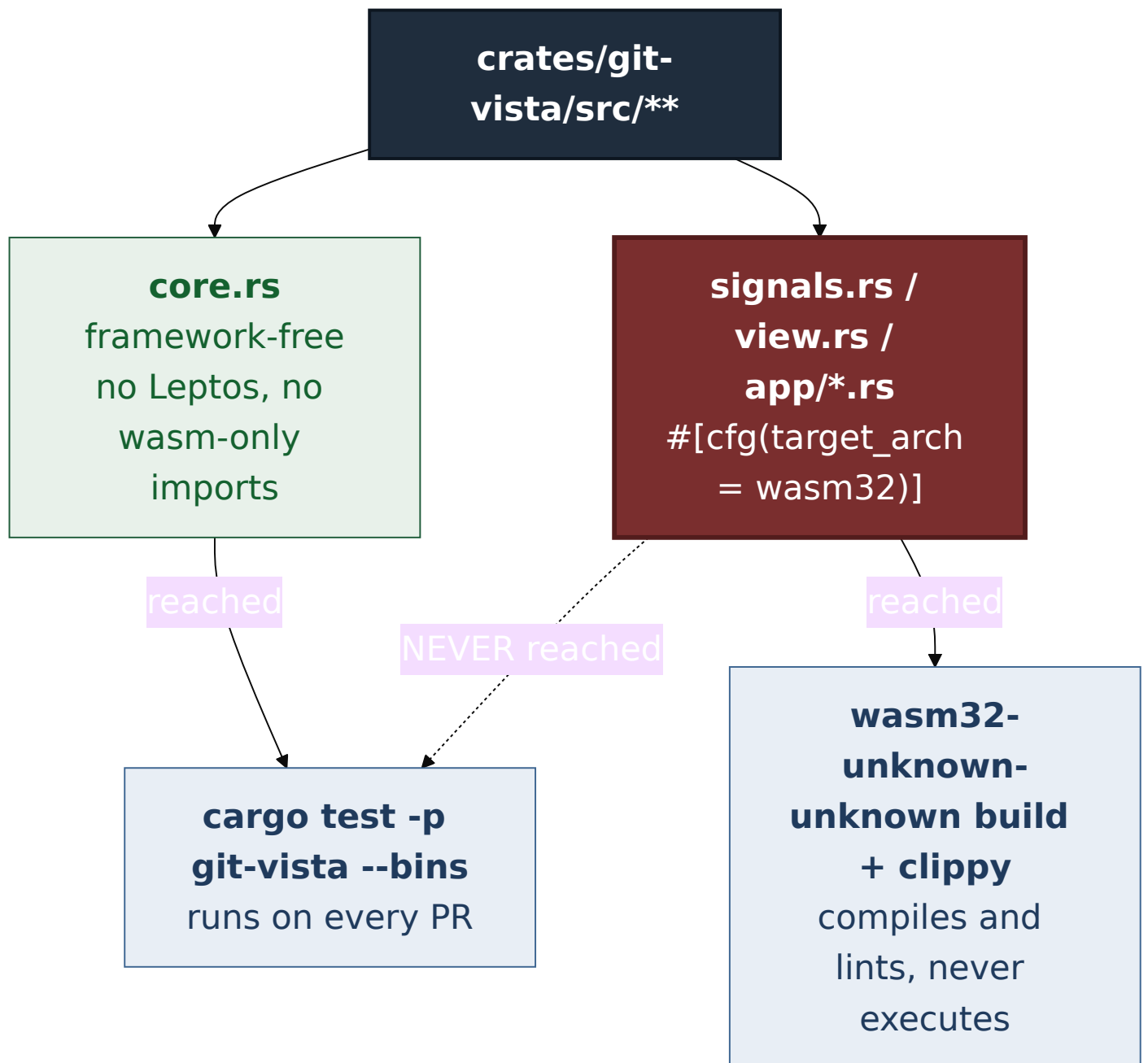
- **Status:** Accepted — implemented, mutation-proved two ways per invariant (19/19 across three PRs)
- **Date:** 2026-09-04
- **Issue:** #612
- **Extends:** —
- **Supersedes / superseded by:** —

Context

`crates/git-vista/src/**` splits into `core.rs` files (framework-free, host-tested) and `signals.rs` / `view.rs` / `app/*.rs` files gated `#[cfg(target_arch = "wasm32")]`. That split is deliberate and old — every `features/*/mod.rs` states it, and it is the shape M1.11's design spec (D1) asked for.

The gap is what CI does with the wasm-only half. It compiles the crate for `wasm32-unknown-unknown` with Trunk. It runs clippy against it. It does neither of those things badly. What it does not do, and cannot be made to do without a browser test runner this repo does not have, is **execute** a single line inside a `#[cfg(target_arch = "wasm32")]` module. `cargo test -p git-vista --bins` — the host suite, the one that runs on every PR — never touches that code at all.

So a decision placed in a wasm-only module is compiled, linted, and unprovable by construction. Not untested by oversight — untestable by the toolchain, on this repository, as configured. A census taken against three sibling issues (#515, #518, #594) found several such decisions sitting exactly there, each one already having caused or nearly caused a real defect. This ADR is the record of moving three of them and what moving them found.



Decision

Move decision logic out of wasm-only modules into a sibling `core.rs`, one seam at a time, keyed off a census rather than a sweep. Each move:

1. Extracts the decision into a pure function in `core.rs`.
2. Replaces the wasm-only call site with a call to that function — never a silent re-derivation.
3. Host-tests the function directly, including the specific invariant that was previously unprovable.
4. Pins the wasm-only call site itself with a source-level census (`include_str!` over the wasm-only file, the same technique `features::ally::audit` already used for markup no host test can mount), so a later change that re-inlines the decision goes red even though it cannot be reached by execution.
5. Mutation-proves the new function two ways, failing on **different** assertions — not two mutations that happen to redden the same test, which is one mutation reported twice.

Three seams moved this way, in order:

PR	Issue rank	What moved	Host tests
#640	—	StashNotice::from_write / from_pop , DrawerBusy (#515, #518)	773 → 781
#642	#2	send() 's OperationKind → endpoint table, persist_if_remote_op , the reattach budget rule	773 → 784
#645	#3, #4	app/mod.rs 's HistoryPhase + its three transition rules; dialogs/confirm.rs 's preview composition	792 → 814

#640 — the stash notice and busy map

StashNotice::from_write / from_pop and DrawerBusy were pure types living in features/stash/signals.rs , wasm-only for no reason but placement — moving them cost a pub use re-export, nothing else. What that move made testable was #515's rule (a lost write reply may only claim Unknown , never a success or a refusal it never observed) and #518's rule (the busy map is keyed on the entry's OID, not its selector, so two writes racing on selectors that momentarily collide cannot unlock each other's row). Both had been *review findings*, per the originating census — found by a person reading the diff, not by a test, because no test could reach them.

#642 — the write-routing table

This is the literal shape of #594: send() matched on OperationKind and called a different api:: function per arm, entirely inside features/operations/signals.rs . A variant swapped for its type-compatible sibling — DiscardTrackedPaths routed to delete_untracked_paths_request , say — compiles clean and sends the wrong git command, with no test able to see it happen. write_route now owns that table in core.rs , and sends_dispatch_matches_the_route_table 's census pairs each OperationKind arm in signals.rs with the api:: call that follows it **in source order**, because counting method names proves nothing when both names appear in the file regardless of which arm reaches which. Two more decisions moved alongside it because they were small once the module was open: the InFlightRemoteOp persistence mapping (round-trip-proved against core 's own reverse mapping) and the #218-adjacent reattach-budget rule, which turned out to be two genuinely different triggers sharing one Settlement value — unified into lost_contact_settlement() , kept structurally distinct as reattach_step .

#645 — the history phase and the sharpest finding of the three

`app/mod.rs` 's `HistoryPhase` and the three effects that move it (`phase_for_epoch_bump`, `promote_seed`, `seed_retry_still_wanted`) moved the same way as the other two. The finding worth stating plainly is in `dialogs/confirm.rs`, and it is the instance #612's own issue body names.

`preview_subject` (in `features::dialogs::core`) and `previewable` (in `features::preview::core`) were **both already** framework-free and host-tested before this ADR. #594 gave them a two-way mutation proof — one mutation removing the variant discrimination, one weakening the default arm — and both came back `caught`. The proof's own report said "both caught."

That report was true and it was not enough. The line that **composed** the two functions —

```
match previewable(preview_subject(&op)) {
    Some(operation) => preview.start(operation),
    None => preview.clear(),
}
```

— lived in `dialogs/confirm.rs`, wasm-only, unreachable by the host suite the mutation proof ran against. Swap the two arms of that `match`, or drop the `None` case, and every existing test — including both #594 mutations — stays green. **A mutation proof over two correctly-tested functions is not a proof of the code that composes them, and "both caught" is a report about the functions, not about the wire connecting them to anything a person actually sees.**

`preview_action` now makes that composition inside `features::preview::core`, returning a two-armed `PreviewAction` (`Start` / `Clear`) rather than a `bool`, because collapsing "no dialog is open" and "this dialog has no picture" onto the same instruction was itself part of the decision, not incidental to it. `confirm.rs` is reduced to plugging the two arms in, and

`the_confirm_dialog_does_not_have_the_two_arms_the_wrong_way_round` reads `confirm.rs` back as text and pairs each `PreviewAction` arm with the **first** `preview.` call that follows it in source order — the same counting problem #642's route-table census solved, for the same reason.

One regression surfaced while building this seam, caught by a guard already in the repo rather than by anything written for this ADR: an early draft passed `preview_subject` to `.map()` as a function value, and

`reachability_census` failed immediately — a function reachable only as a value has no statement-shaped call site and reads as dead code. The fix was spelling the call explicitly in both arms of the caller's `match`. This is recorded because it is evidence for the enforcement decision below, not because it was hard to fix.

Enforcement — what actually catches the next one

The issue asks, in substance, "what stops this from happening again." Three options were on the table.

Not a clippy lint. Clippy lints syntax and types. "This value is a decision the rest of the app depends on" is neither. A wasm-only module is *saturated* with functions that are pure by signature — every `#[component]` body, every markup-composing helper, every closure passed to `create_effect`. A lint keyed on "pure function in a `#[cfg(target_arch = \"wasm32\")]` module" would fire on nearly everything in `dialogs/`, `menu.rs`, and `render/`, and a lint tuned down until it stopped would have stopped firing on the case that mattered too. The same argument kills "grep wasm-only modules for pure functions" as a standalone mechanism — it is the same lint, run by hand.

What is already working, and should be leaned on rather than rebuilt: `reachability_census`. It did not exist for this ADR; it caught this ADR's own regression anyway, because "a core function with no real call site" is a mechanically checkable fact in a way "this logic is a decision" is not. Every seam moved in this ADR kept its host-tested function reachable *because* that census would have failed otherwise. That is real leverage and it is already paid for.

What is worth building, honestly scoped: an inventory, not a verdict. A new host test enumerates the `#[cfg(target_arch = \"wasm32\")]` module list (from `main.rs`'s and each `features/*/mod.rs`'s `#[cfg]` declarations) and the set of `include_str!` targets across the existing wasm-only-reading host tests (`ally::audit`, `offline_guard_audit`, and the per-seam censuses this ADR adds), then asserts every wasm-only module over some line-count floor has at least one host test reading it. This does not check that the *right* decisions moved, or that a moved decision is *correctly* pinned — it checks that nobody has been forgotten, the way #612's original census had to be built by hand once. Making it a standing, always-current test turns a one-time manual audit into a fact that rots loudly instead of silently.

What does the actual enforcing is neither of those. It is the per-seam census this ADR's three PRs each wrote: a specific `include_str!` reading a specific wasm-only file, asserting a specific call is made in a specific arm. That is not a mechanism installable once — it is a habit, written at the moment a decision moves, argued in the module's own doc comment the way `features::preview::core`'s doc on `DialogSubject` now argues why the vocabulary stays narrow. `features/mod.rs`'s header states the `core.rs / signals.rs` split; a line naming this expectation belongs there too, so the next person moving a seam finds the pattern instead of reinventing it.

The honest summary: the inventory test enforces **visibility** — that a gap cannot go unnoticed the way #612's did. It does not enforce **correctness** — that requires the per-seam census, written by a person who understands the specific decision, every time.

Alternatives considered

- **A wasm test runner** (`wasm-pack test --headless`, or similar) would let the host suite genuinely execute wasm-only code and remove the need for text-census workarounds entirely. Rejected for this ADR: it is infrastructure, not a decision, sized well past what three move-slices warrant, and the existing browser-driven Playwright suite already covers some of the same ground at the DOM level (imperfectly — see #623/#629 on that suite's own reliability). Worth its own issue if the pattern in this ADR keeps recurring past the modules already named.
- **Move everything wasm-only can reach into core, unconditionally.** Rejected as a category error: markup composition, DOM event wiring, and `create_effect` closures that only sequence signal reads/writes are not decisions — moving them buys nothing and would bloat `core.rs` files with code that has no invariant to pin. The census-driven, one-seam-at-a-time approach exists specifically to separate "this is a decision" from "this merely runs in a wasm-only file."
- **A single shared mutation-proof harness across all three seams** was considered and rejected: each seam's invariants are different enough (an epoch-comparison guard, a route table, a composed `Option` chain) that a shared harness would either under-specify or force an abstraction with one real caller — the anti-pattern this repo's own CLAUDE.md style already warns against.

Consequences

- Host test count for `git-vista` moved 773 → 814 across the three PRs, with zero behaviour change in any of them — every move was re-export-and-call, never re-derive.
- `reachability_census` and the per-seam text censuses are now load-bearing in a way they were not designed to be for this issue specifically; a future refactor that deletes a census comment without reading why it exists reintroduces exactly the gap this ADR closes.
- The inventory test described above is **not yet built**. It is scoped as the next piece of work under #612, tracked separately from this ADR.
- `canvas.rs`'s 409 handler still sets `HistoryPhase::DriftReloading` directly, in wasm-only code. #645's own census (`every_phase_the_shell_can_set_comes_from_a_decision_this_file_makes`) names this gap in its own assertion message rather than implying the move is complete. It is not disqualifying — `DriftReloading`'s only content is which epoch it is reloading into, not a branching decision — but it is the one variant this ADR's move did not fully close, and a future slice should either move it or record why it is exempt, the way #515/#518's `signals.rs` docstring records what still lives there and why.

Since this ADR was written

The two bullets above that describe unfinished work are now history, not status. They are left as written — the record of what was true when the decision was made — and superseded here.

The inventory test is built. #649 landed it as `crates/git-vista/src/wasm_module_census.rs`, scoped exactly as this ADR argued: it enumerates the `#[cfg(target_arch = "wasm32")]` module set by scanning `main.rs` and each `features/*/mod.rs`, collects every `include_str!` target, and names any wasm-only file at or above 150 lines that no host test reads and no argued `EXEMPT` entry covers. It enforces **visibility**, never correctness — its own module doc says so at length, and lists five separate things it does not prove.

The six gaps it named are closed. In the order they went:

File	Closed by	What moved
app/canvas.rs	#650	the 409 drift response, and the ordering rule that ties the announcement to the epoch the bump produced
features/shell/signals.rs	#652	the overlay payload map and Overlay::ALL 's completeness invariant
print.rs , render/labels.rs	#653	the GitHub link rules, the ref glyph mapping, the badge palette — one copy each, in two files
gestures.rs , features/diff/staging_view.rs	#653	the canvas shortcut map with its three ordering rules, and the roving-row key map both files held a copy of

Two things that only became visible once the census existed, and that the "what actually enforces this" section above did not anticipate:

- **A duplicated decision is one gap counted twice, and the census counts it once per file.** print.rs and render/labels.rs each held their own copy of the badge glyph mapping and the badge palette, each carrying a comment promising it matched the other. Neither file's EXEMPT entry mentioned the other. The census cannot see duplication — it asks only whether *someone* reads a file — so the pairing had to be noticed by a reader. Where a move finds a duplicate, the destination should name the one intended divergence explicitly rather than leaving two copies that happen to agree: BadgeSurface::Screen / Paper exists for exactly that, and a test asserts HEAD is the *only* kind that differs.
- **A wasm-only file can contain a #[cfg(test)] mod tests that has never run.** print.rs carried three assertions about its commit-link rule. The module around them is #[cfg(target_arch = "wasm32")], so the host build compiled none of it: no failure, no pass, silence — the exact shape this ADR is about, sitting inside a file that looked tested. The census could not see it either, since the file was EXEMPT for unrelated reasons. #653's suite now asserts print.rs contains no mod tests at all, with the reason in the message. **Any wasm-only module is worth grepping for mod tests before its coverage is believed.**

One EXEMPT entry remains that names real debt:
dialogs/open_url.rs . The rest of the table is argued-thin files whose decisions already live in a host-tested core .

Signed: max · 2026-09-05